TARGETING SPRINT 2 - COUNTER-AIR EVALUATION - CATEGORIES 1, 2, 3 & 4 (INTEGRATED MOSA C2 / PROJECT AVO SAFETY INTERLOCK & 10 U.S.C. 4023 SUBSYSTEM PAYLOAD)

Solicitation: DIU / DRPM UxS Suitable Warfighting Adaptive Payloads (SWAP)-USV Sprint 2 (10 U.S.C. 4025 / 4022 / 4023)

Submitting Entity: Humphrey Virtual Farms LLC (701 E Drew, Madill, OK 73446-2620 | 100% U.S. Sovereign Small Business, 0% FOCI)

Statutory Identifiers: CAGE Code: 1AHA8 | SAM.gov UEI: S1M4ENLHTDH5 | Non-Traditional Defense Contractor (10 U.S.C. 4022)

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Public Verification Repository: https://github.com/mrshumphrey3251-ai/hvf-media-matrix-public

Shared Core C2 Cryptographic Merkle Root (SHA-256): d65049efcf06298f84d41436131794fc3cc5607873a2a596b5ad355f6ee67ccb

Cross-Referenced Active Filings: DoD CDAO Tradewinds Docket 10-26-4046 (Status: Compliant/Queued for Assessment -- Oct 29, 2026), U.S. Army DEVCOM ARL BAA W911NF-23-S-0001 ($750K White Paper Transmitted), USACE ERDC BAA ($1.2M), and DHS S&T LRBAA ($1.0M OIP Confirmed)

1. BLUF: TARGETING SPRINT 2 - COUNTER-AIR EVALUATION - CATEGORIES 1, 2, 3 & 4

Humphrey Virtual Farms LLC (CAGE: 1AHA8 | UEI: S1M4ENLHTDH5) submits Project Ebony v8.0.0 -- a deterministic, zero-cloud, Three-Brain Maritime Autonomous C2, Project AVO Interoperability, and Human-in-the-Loop Weapon Safety Interlock payload architecture -- for immediate integration across Category 1 (Heavy Kinetic AGM-114L/AGM-179 GFI TDP), Category 2 (APKWS Precision Rocketry), Category 3 (Stabilized RWS / Stinger / HPL), and Category 4 (Modular Box/Tube Effectors) USV platforms under 10 U.S.C. 4025 / 4022, as well as direct experimental subsystem/testbed procurement under 10 U.S.C. 4023 ($1,485,000.00 Phase-1 Prototype OTA / Experimental Subsystem Procurement).

[+] Solving the Contested Maritime Cost-Exchange & Safety Interlock Gap: Counter-air USVs operating in Denied, Degraded, Intermittent, and Limited (DDIL) and GNSS-denied littoral environments face catastrophic cyber-physical risks: spoofing of remote firing links, neural target-PID hallucinations in sea clutter, and loss of human-in-the-loop abort custody. Project Ebony v8.0.0 eliminates these failure modes at a 28.5W / 24V DC shipboard power envelope.

[+] Production-Ready Software C2 & SIL/HIL Baseline (640 Verified Modules): Verified across 640 compiled Python C2 modules, 85/85 passing automated test suites, 17/17 blocked Red-Team penetration vectors with automatic single-letter tamper zeroization, and 52 bare-metal SQLite WORM/vault tables on local loopback (127.0.0.1) with 0% commercial cloud dependency.

2. THREE-BRAIN MOSA ARCHITECTURE & PROJECT AVO C2 INTEROPERABILITY

Project Ebony v8.0.0 enforces strict physical and logical separation across three dedicated compute enclaves designed to drop directly into standard 20/40ft ISO containerized USV mission bays and interface natively with DoW's Project AVO (Autonomous Vehicle Orchestrator):

[+] Brain One (Deterministic Effector Safety Interlock & Human-in-the-Loop Custody Core): Executes a <50 microsecond bare-metal safety loop governing fail-safe over-the-horizon remote arming, firing, safe/de-arming, and emergency mission abort across CAT 1-4 launchers. Models a 2.04 microsecond RS-485 Modbus RTU (Function Code 05) emergency actuator disconnect and a 36.75 nanosecond Silicon Carbide (SiC) MOSFET gate cutoff target -- 1,360,544x faster than a legacy 50ms marine PLC -- ensuring instant physical weapon safe/de-arming if human-in-the-loop custody or stabilization envelopes (pitch/roll launch limits) are violated.

[+] Brain Two (DDIL / GNSS-Denied Localization & 360-Degree Clutter-Resilient PID Analyst): Executes zero-cloud edge neural inference for 360-degree airborne threat detection, tracking, and positive identification (Group 1-5 UAS, rotary-wing, and go-fast surface vectors) in high-sea-state clutter, plus optical/inertial dead-reckoning when GNSS is jammed or spoofed. Brain Two communicates with Brain One exclusively across a unidirectional, read-only memory diode with zero physical wire to launcher firing circuits, mathematically preventing adversarial AI spoofing or sensor clutter from triggering unauthorized weapon release.

[+] Brain Three (Sub-Millisecond WORM Merkle DAG Provenance & SIL/HIL Telemetry Arbiter): Computes and seals a cryptographic SHA-256 Write-Once-Read-Many (WORM) Merkle Directed Acyclic Graph (DAG) across 52 local SQLite tables in <0.2 milliseconds per event, logging every Project AVO cueing packet, operator arm/fire/abort command, and 3DOF/6DOF kinematic engagement state for Test Event 1 and Test Event 2 engineering depositions.

3. ALIGNMENT WITH SPRINT 2 DESIRED ATTRIBUTES & TEST EVENTS 1 & 2

[+] Project AVO & 3rd-Party Open C2 Interface (MOSA SDK/API Ready): Provides a documented, modular JSON/REST and RS-485/CAN-bus C2 interface package (SDK/API and emulator) ready for immediate integration with Project AVO, processing offboard tactical cueing coordinates and enforcing human-on-the-loop/human-in-the-loop firing authorization.

[+] SIL/HIL Kinematic Validation & 30-Day Test Event 1 Readiness: Includes a built-in, deterministic Software-in-the-Loop (SIL) and Hardware-in-the-Loop (HIL) simulation harness modeling target kinematics, launch platform stabilization limits, and effector error conditions, ready for live onsite demonstration within 30 calendar days of down-selection (Test Event 1) and Little Creek, VA operational integration (Test Event 2).

[+] SWaP-C & Containerized Logistics Affordability: Operates at 28.5W continuous auxiliary draw on native 24V DC marine vessel power (requiring zero specialized MHE or shore infrastructure), compatible with C-130/C-17 air transport, commercial highway trailering, and 20ft/40ft ISO containerized well-deck deployment.

[+] Dual Acquisition Readiness (Teamed CAT 1-4 Integration + 10 U.S.C. 4023 Experimental Procurement): Available immediately to DIU and DRPM UxS either (a) paired with DIU down-selected CAT 1-4 USV hull and launcher primes as the standardized MOSA C2 / Human-in-the-Loop Safety Interlock & Project AVO bridge, or (b) procured directly under 10 U.S.C. 4023 as an experimental C2 safety-interlock payload and SIL/HIL test article ($1,485,000.00 / 12-Month Fixed-Price Schedule; 100% Background IP retained under DFARS 252.227-7018; 100% Section 889 NDAA compliant).